

A graph from its inverse

$$f(x) = 2^x \quad g(x) = \log_2 x$$

Which points on g correspond to (0, 1), (1, 2), and (2, 4) on f?

Domain and range

$$g(x) = \log_2 x$$

What are the domain and range?

A logarithmic asymptote

$$g(x) = \log_2 x$$

What vertical line is the asymptote?

A logarithmic intercept

$$g(x) = \log_2 x$$

Where does the graph cross the x-axis?

Increasing or decreasing

$$g(x) = \log_{1/3} x$$

How does the graph change as x increases?

Approaching the domain boundary

$$g(x) = \ln x$$

What happens to $g(x)$ as positive x approaches 0?

A horizontal shift

$$g(x) = \log_2 (x - 4)$$

What are the domain and vertical asymptote?

A vertical shift

$$g(x) = \log_2 x + 3$$

Where does the parent point (1, 0) move?

A vertical reflection

$$g(x) = -\ln x$$

What are the intercept and direction of change?

A horizontal reflection

$$g(x) = \ln(-x)$$

What are the domain and vertical asymptote?

A vertical stretch

$$g(x) = 3\log_2 x$$

Where does the parent point (2, 1) move?

A scaled input

$$g(x) = \ln(2x)$$

What is the domain?

A reversed input

$$g(x) = \log_3 (5 - x)$$

What are the domain and vertical asymptote?

A transformed graph

$$g(x) = -2\log_3(x + 1) + 4$$

What transformations produce this graph from the parent graph?

An x-intercept after shifting

$$g(x) = \log_2 (x - 1) - 3$$

Where does the graph cross the x-axis?

Two input restrictions

$$g(x) = \ln(x - 2) + \ln(5 - x)$$

What is the domain of the entire expression?

A logarithm of a quadratic

$$g(x) = \ln(x^2 - 9)$$

What intervals make the argument positive?

A graph from a point and asymptote

The vertical asymptote is $x = 3$, and the graph passes through $(4, 2)$.

$$g(x) = \log_2 (x - h) + k$$

What values of h and k fit these features?

Sketching a shifted logarithm

$$g(x) = \ln(x + 2) - 1$$

What sketch shows its domain, intercepts, and asymptote?

An inverse with a restriction

$$f(x) = e^{x-2} + 3$$

What are the formula and domain of the inverse function?